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Mobile device accessory

Abstract

An accessory case for a mobile device includes a housing, a channel, and an input board. The housing has a base wall and a perimeter wall extending away from the base wall, which together define a cavity to receive a mobile device. The channel is formed within the base wall, and extends inward from but not through the base wall to define a pocket that is separate from the cavity. The input board is received within the pocket. The input board is formed by folded segments that have one or more inputs. The input board is configured to transition between a folded position and an unfolded position. In the folded position, a width of the input board is less than a width of the channel. In the unfolded position, the width of the input board is greater than the width of the channel.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of International Patent Application No. PCT/US2021/024857, filed Mar. 31, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application No. 63/002,754, filed Mar. 31, 2020, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Phones and other mobile devices are frequently used to input and send messages, notes, and other media to different devices on a network. Although convenient, mobile devices typically lack inputs that enable efficient and fast data input. Further, because data input can be inefficient, mobile devices frequently do not contain certain software classes (e.g., word processing programs, spreadsheets, etc.) that are often found on traditional computers.

SUMMARY

(2) One exemplary embodiment relates to an accessory case for a mobile device. The accessory case includes a housing, a channel, and an input board. The housing has a base wall and a perimeter wall extending away from the base wall. The perimeter wall and the base wall together define a cavity to receive a mobile device. The channel is formed within the base wall. The channel is defined by walls that extend inward from a first perimeter side of the base wall, toward but not through a second perimeter side of the base wall. The channel defines a pocket within the base wall that is separated from the cavity. The input board is received within the pocket. The input board is formed by several folded segments that support several inputs. The input board is configured to transition between a folded position and an unfolded position. In the folded position, a width of the input board is less than a width of the channel. In the unfolded position, the width of the input board is greater than the width of the channel.

(3) Another exemplary embodiment relates to a mobile device. The mobile device includes a cover, a display, and an input board. The cover is formed from a substantially flat rear wall and a side wall extending upwardly away from the rear wall. The side wall extends at least partially around the rear wall. The rear wall and the side wall together define a cavity that receives a rechargeable battery. The display extends across a portion of the cover to enclose the rechargeable battery. The display is configured to present a graphical user interface that is powered by the rechargeable battery. The input board is coupled to the cover and extends within a recess formed within and along at least a portion of the rear wall. The input board is formed by several folded segments having inputs that are configured to interact with the graphical user interface. The input board is configured to transition between a folded position and an unfolded position. In the folded position, the input board is received entirely within the recess so that the entire keyboard is positioned inward from an outermost surface of the rear wall. In the unfolded position, the input board is removed from the recess and is defined by a width that is greater than a width of the cover.

(4) Another exemplary embodiment relates to a mobile device. The mobile device includes a cover, a display, and an input board. The cover is formed from a substantially flat rear wall and a side wall that extends upwardly away from the rear wall. The side wall extends at least partially around the rear wall. The rear wall and side wall together define a cavity that receives a rechargeable battery. The display extends across a portion of the cover to enclose the rechargeable battery. The display is configured to present a graphical user interface that is powered by the rechargeable battery. The input board is coupled to the cover and extends within a recess formed within and along at least a portion of the rear wall. The input board is formed from several folded segments that support inputs that are configured to interact with the graphical user interface. The input board is configured to transition between a folded position and an unfolded position. In the folded position, the input board is defined by a width that is less than a length of the rear wall. In the unfolded position, the input board is defined by a width at least 50 percent larger than the length of the rear wall.

(5) The invention is capable of other embodiments and of being carried out in various ways. Alternative exemplary embodiments relate to other features and combinations of features as may be recited herein.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:
- (2) FIG. 1 is a front view of a mobile device within a case, according to an exemplary embodiment;
- (3) FIG. 2 is a rear view of the mobile device of FIG. 1, depicting an accessory in a first configuration;
- (4) FIG. 3 is a perspective view of the mobile device of FIG. 1, depicting the accessory of FIG. 2 in a second configuration;
- (5) FIG. 4 is a perspective view of the mobile device of FIG. 1, depicting the accessory of FIG. 2 in a third configuration;
- (6) FIG. 5 is a side view of the mobile device of FIG. 1, depicting a mobile device stand that can be formed in the accessory;
- (7) FIG. 6 is a top view of the accessory in a fully unfolded position, depicting the hinge lines in the accessory;
- (8) FIG. 7 is a side view of the mobile device of FIG. 1, depicting an alternative device stand positioned on the case; and
- (9) FIG. 8 is a top view of another accessory in a fully unfolded position, depicting another configuration of foldable panel sections.

DETAILED DESCRIPTION

- (10) Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.
- (11) Referring to the FIGURES generally, the various exemplary embodiments disclosed herein relate to keyboard and/or input systems and apparatuses that can be used with a mobile device and/or a mobile device case. Specifically, the mobile devices are equipped with a thin, full keyboard or input board that can be selectively deployed to improve the functionality of a standard phone or mobile device. The keyboard can be folded and stored in a compact shape along or within the mobile device case, which allows the keyboard to be easily transported with the mobile device. If a user wishes to access the keyboard, the keyboard can be released or otherwise removed from within the mobile device case. Once partially extended from the mobile device case, the keyboard can be unfolded to reveal a full keyboard that can then be used to input text into the mobile device. The keyboard can further include a mounting tab that can be folded to prop the mobile device and mobile device case in an orientation that is conducive to typing with the keyboard. By providing a full keyboard, various different programs (e.g., word processing programs, emails, etc.) that might normally be reserved for traditional computers or laptops can be effectively used with a standard mobile device (e.g., a smart cellular phone). Similarly, the speed with which a user can enter information is improved relative to conventional mobile devices.
- (12) As shown in FIG. 1, a mobile device **10** is depicted. The mobile device **10**, which is depicted as a smart cellular phone, can take several different forms, including a tablet, computer, smart watch, laptop, etc. that is configured to be transported to various locations. The mobile device **10** can be configured to communicate over various different communication frequencies, including cellular, Wi-Fi, Bluetooth, and the like. Similarly, the mobile device **10** can be configured to support one or more of video calls, voice calls, text messaging (SMS), internet messaging, and/or e-mail, for example, which can be communicated over and through various different networks.
- (13) With additional reference to FIG. 2, the mobile device **10** generally includes an outer shell or structure that is formed from a cover **12** and a display **14**. The cover **12** is formed from a substantially flat rear wall **16** and a side wall **18** that extends at least partially around the rear wall

16. The offset between the top of the side wall **18** and an inner surface of the rear wall **16** together define a cavity **20**. The cavity **20** receives the various different electronic components that power the mobile device, including for example a rechargeable battery **22**, an antenna (not shown), and one or more processors, transistors, chips, and semi-conductors. In some examples, the rechargeable battery **22** is completely enclosed within the cover **12** and display **14** so that the rechargeable battery **22** is protected from external contaminants (e.g., moisture, dust, etc.). In some examples, one or more terminals **24** are formed within or through the cover **12** to form externally accessible connections that can be used to transmit and/or receive electrical power between the rechargeable battery **22** and an external power source (e.g., a wall charger) or consumption device (e.g., an accessory, as explained in additional detail below).

(14) The display **14** is coupled to the cover **12**, and is configured to present a wide variety of data to a user through a graphical user interface (GUI) **26**. The GUI **26** can be configured to present messages, word processing programs, articles, spreadsheets, games, and other interactive items that can be adjusted or otherwise interacted with using a variety of inputs on the mobile device **10**. The inputs can be presented as part of the GUI **26**, in a touch screen, or can be physical buttons presented on and supported by the cover **12** and/or the display **14**. In some examples, the inputs are a series of alphanumeric keys that can be used to type different messages into the GUI **26**. Various other types of inputs can be formed within the cover **12** and/or display **14**, including power toggle switches, volume adjustments, and the like. In some examples, the various inputs include both physical actuators and touch screen actuators, both of which are configured to interact with the GUI **26** on the display **14**. The display **14** can be a liquid crystal display (LCD), light emitting diode (LED) display, or other similar suitable type of solid state light source display, for example. As discussed above, the display **14** can be powered by the rechargeable battery **22**.

(15) Although convenient, the size and location of the various inputs on the cover **12** and/or display **14** may negatively impact the speed and accuracy with which a user can create and relay messages. For example, the narrow spacing between inputs on a touch screen may result in frequent typographical errors and accidental inputs that are preferably avoided. Similarly, the spacing between the inputs significantly hampers the speed at which a user can enter information, as typically only one or two fingers can be used at a time to interact with different inputs. These data entry difficulties caused by space constraints are such that conventional mobile devices **10** often omit word processing programs and other frequently used software.

(16) To address the data entry issues, the mobile device **10** includes an accessory, shown as input board **30**. The input board **30** is formed of a series of folded segments that support different inputs (e.g., keys, buttons, trackpads, etc.) that can be used to input data into the mobile device **10**. The folded segments are configured to selectively deploy to transition the input board **30** between a folded position (e.g., the position shown in FIGS. 2-3) and an unfolded position (e.g., the position shown in FIGS. 4 and 6). The input board **30** is configured to provide a faster and more efficient mechanism for data entry than the other inputs positioned on either of the cover **12** or the display **14**.

(17) With specific reference to FIGS. 4 and 6, the input board **30** is shown in additional detail. The input board **30** is generally formed from a first, main panel section **32**, a second panel section **34**, and a third panel section **36** that support a variety of inputs **38**. The variety of inputs **38** can include alphanumeric keys **40** and one or more track pads **42**, for example, that are spaced about the first panel section **32**, second panel section **34**, and third panel section **36**, respectively. In some examples, the keys **40** can be electronic and programmable. The keys **40** may be spring actuated, such that the individual keys **40** compress vertically into the input board **30** when pressed, and return to their level position when not pressed. In some examples, the keys **40** each extend upward from the panel sections **32**, **34**, **36** by a distance that is greater than a thickness of the panel sections **32**, **34**, **36**. In still other examples, the keys **40** are touch or pressure sensitive, and do not move when pressed. The input board **30** can include backlighting, as well as other control keys, such as

volume control, power control, and the like.

(18) The track pad **42** can be pressure and/or touch sensitive, and can be formed as a rectangle or square area. In some examples, the track pad **42** can be depressed to make selections on the mobile device **10** as well. Although shown on a right side of the input board **30** in FIG. **6**, the track pad **42** can also be positioned at or near the bottom center of the first panel section **32** as well. The track pad **42** can also be formed at or near an edge of one of the panel sections **32**, **34**, **36**, and one of the panel sections **32**, **34**, **36** contains all of the alphanumeric keys **40**. In some examples, the track pad **42** extends across multiple panel sections **32**, **34**, **36**. In some examples, the track pad **42** includes selection buttons that operate similar to a standard mouse. The selection buttons can be positioned along a bottom or top of the track pad **42**, for example. In still other examples, the track pad **42** can include a touch sensitive button or ball. The input board **30** can be formed from a flexible polymeric material, for example, that can be pliable to bend when being manipulated or moved from the case **110** or cover **12**.

(19) The input board **30** is designed to be compact and easily storable within either the cover **12** or a case **110** of the mobile device **10**. To easily store the input board **30**, the panels **32**, **34**, **36** that form the input board **30** are arranged to fold. As depicted in FIG. **4**, for example, the first panel section **32** is connected to the second panel section **34** and the third panel section **36** along fold lines **44**, **46**. In some examples, the first and second fold lines **44**, **46** extend approximately parallel (e.g., ± 15 degrees) of each other. The second panel section **34** and third panel section **36** are configured to fold relative to the first panel section **32** about the first and second fold lines **44**, **46**, which in turn transitions the input board **30** between the folded, storage position and the unfolded, deployed position for use. In some examples, the fold lines **44**, **46** do not include any inputs (e.g., keys **40** or trackpads **42**). The fold lines can be formed of flexible materials, while the panels **32**, **34**, **36** can be formed of a more solid material. In still other embodiments, the input board **30** includes hinges along the fold lines **44**, **46** to help promote the movement of the second panel section **34** and third panel section **36** relative to the first panel section **32** along the fold lines **44**, **46**.

(20) In the folded position, the second panel section **34** and third panel section **36** extend along a front surface of the first panel section **32**, as depicted in FIG. **3**. In some examples, the second panel section **34** and the third panel section **36** are each configured to be about half the size of the first panel section **32**. Accordingly, when the second panel section **34** and the third panel section **36** are folded inward, into the storage position, the outer perimeter of the input board **30** is approximately the same size as the outer perimeter of the first panel section **32**. Various other folding configurations for the input board **30** can be used as well. For example, the second panel section **34** and third panel section **36** can each be designed to have approximately the same width as the first panel section **32**. Similarly, the folding sequence of the input board **30** panel sections **32**, **34**, **36** can be interchangeable or otherwise chosen to promote a distinct and consistent folding pattern. The panel sections **32**, **34**, **36** can be adjusted and sized to make the folding sequence intuitive and readily apparent to a user. Differently sized hinges can be used based upon the fold types and patterns used by the input board **30**. In some examples, the input board **30** includes one or more magnets that are configured to secure the folded panel sections **32**, **34**, **36** in place. In still other examples, mechanical or electro-mechanical locks can be used to selectively secure the panel sections **32**, **34**, **36** in the folded orientation.

(21) In some examples, and as depicted in FIG. **6**, the input board **30** includes additional mounting panel sections that can be folded to prop up the mobile device **10** for use with the input board **30**. For example, the input board **30** can include a fourth panel section **48** and a fifth panel section **50**. In some examples, neither of the fourth panel section **48** or the fifth panel section **50** include inputs. The fourth panel section **48** and fifth panel section **50** extend upwardly away from a top side of the first panel section **32**. The fourth panel section **48** is connected to the first panel section **32** along a third fold line **52**. The third fold line **52** extends collinear with a top edge of the first panel section

32, approximately perpendicular (e.g., ± 15 degrees) to the first fold line and second fold line **44**, **46**. The fifth panel section **50** is connected to the fourth panel section **48** along a fourth fold line **54**. The fourth fold line **54** extends approximately parallel to the third fold line **52** and approximately perpendicular to the first fold line **44** and the second fold line **46**. As depicted in FIG. 6, the fourth panel section **48** and the fifth panel section **50** together are defined by a width that is approximately equal to the width of the first panel section **32**. In some examples, the fifth panel section **50** is defined by a width that is different than a width of the fourth panel section **48**. For example, and as shown in FIG. 6, the width of the fifth panel section **50** can be larger than the width of the fourth panel section **48**. In the folded position, the fourth panel section **48** and the fifth panel section **50** are folded backward, and extend along a rear surface of the first panel section **32**. Accordingly, when the input board **30** is in the fully folded position, the input board is arranged in three stacked layers, the lowermost of which is the fourth panel section **48** and the fifth panel section **50**. The first panel section **32** forms the middle layer. The second panel section **34** and third panel section **36** form the top layer of the input board **30**. In some examples, each of the panel sections **32**, **34**, **36**, **48**, **50** is defined by a thickness that is less than $\frac{1}{3}$ of the thickness of a thickness defining the mobile device **10**. The thickness of the panel sections **32**, **34**, **36**, **48**, **50** can be less than $\frac{1}{4}$, less than $\frac{1}{6}$, less than $\frac{1}{8}$, less than $\frac{1}{10}$, less than $\frac{1}{20}$, less than $\frac{1}{30}$, or less than $\frac{1}{40}$ of the thickness of the mobile device **10**, for example. In still other examples, the panel sections **32**, **34**, **36** are separate panels that can be arranged into the full input board **30** once removed from the cover **12** or case **110**.

(22) The input board **30** is configured to be deployed for use with the mobile device **10**. To deploy the input board **30**, the input board **30** must first be removed from its storage position on or along the mobile device **10**. As will be explained in additional detail below, the input board **30** can be stored within or along the cover **12** of the mobile device **10** or can be positioned or otherwise received within a protective case **110** that is received around the mobile device **10**. The input board **30** can be fully removable and detachable from the mobile device **10** and/or case **110**, or can be hingedly or otherwise coupled to the mobile device **10** and/or case **110** so that the deployment process begins by rotating the input board **30** relative to the mobile device **10**.

(23) Upon removing the input board **30** from the storage position on or along the mobile device **10** or case **110**, the input board **30** can first be activated or otherwise provided with electrical power. In some examples, the input board **30** has its own onboard power source (e.g., batteries, etc.). Once removed from storage, the input board **30** can be powered on using a switch or by pressing an input on the input board **30**, for example.

(24) In some examples, the input board **30** is configured to draw electrical power from the rechargeable battery **22** within the mobile device **10**. Accordingly, the input board **30** and cover **12** (or case **110**) can include externally accessible terminals **56**. The externally accessible terminals **56** can extend through the cover **12** or case **110** to provide a selective coupling location for the input board **30**. Likewise, one or more of the panel sections of the input board **30** can support both data terminals **58** and power terminals **60**. The data terminals **58** and power terminals **60** are configured to be placed into contact with the externally accessible terminals **56** to create an electrical connection between the input board **30** and the mobile device **10**. Electrical power from the rechargeable battery **22** can be transmitted through the power terminals **60**, while uni-directional or bi-directional communication can occur between the input board **30** and the mobile device over the data terminals **58**. The data terminals **58** can be configured to transmit signals that correspond with the detected selections (e.g., the inputs pressed or otherwise actuated) made by a user. Accordingly, inputs selected on one or more of the panel sections of the input board **30** can be communicated to the mobile device **10**, where these selections can be used to interact with or otherwise manipulate the GUI **26** on the display **14**. In some examples, the data terminals **58** and the power terminals are positioned in a cluster along the first panel section **32**. Alternatively, the data terminals **58** and the power terminals **60** can be supported by the fifth panel section **50**. In still other examples, power

terminals **60** are positioned on the first panel section **32** while the data terminals are positioned on the fifth panel section **50**. When the terminals **58, 60** are positioned on different panel sections (or depending on the location where the input board **30** is stored), the terminals **58, 60** may be automatically disconnected from the externally accessible terminals **56** whenever the input board **30** is in the fully folded position or whenever the input board **30** is in the storage position within the cover **12** or the case **110**. In still other examples, the input board **30** and the mobile phone **10** have a constant wired connection that is uninterrupted regardless of the position of the input board **30** relative to the mobile phone **10**.

(25) The input board **30** can be deployed by first creating a coupling between the terminals **56** upon the mobile device **10** and/or case **110** and the terminals **58, 60**, then unfolding the different panel sections of the input board **30**. For example, and as depicted in FIG. 3, the mobile device **10** is coupled to the at least partially folded input board **30**. In some examples, one or more of the panel sections can include magnets (not shown) that help guide the terminals **56** into contact with the terminals **58, 60** to create the electrical connections that allow the input board **30** to operate. In some examples, and explained in additional detail below, the coupling formed between the input board **30** and the mobile device **10** is not readily removable, and the input board **30** remains hinged to the mobile device **10** when in both the storage position and the deployed position. Accordingly, a wired connection can be maintained between the input board **30** and the mobile device **10** at all times.

(26) With the electric coupling established between the terminals **58, 60** and the externally accessible terminals **56**, the input board **30** can be unfolded. In some examples, the input board **30** is first rotated about an axis that is at least partially defined by an outer surface of the cover **12** or an outer surface of the case **110**. In other examples, the input board **30** rotates about an axis defined by a hinge that is positioned within or upon the cover **12** of the mobile device **10**. As depicted in FIG. 3, the input board **30** can rotate to a position where the input board **30** and the display **14** of the mobile device **10** form an obtuse angle, as depicted in FIG. 5.

(27) With the input board **30** and mobile device **10** positioned, the remaining portions of the input board **30** can be unfolded. As depicted in FIGS. 3 and 4, the second panel section **34** and third panel section **36** can each be unfolded from their respective resting positions along the first panel section **32**. The second panel section **34** can be rotated outwardly about the first fold line **44**, until the second panel section **34** extends approximately parallel and coplanar with the first panel section **32**. Similarly, the third panel section **36** can be rotated outwardly about the second fold line **46**, until the third panel section **36** extends approximately parallel and coplanar with the first panel section **32**. With both the second panel section **34** and third panel sections **36** extending away from the first panel section **32** to provide a full array of alphanumeric inputs **40** and one or more trackpads **42**. Although described as including three panel sections **32, 34, 36** supporting inputs, various embodiments of the input board **30** can include two or less panel sections or four or more foldable panel sections.

(28) In some examples, the input board **30** is configured to prop up or support the mobile device **10**, as depicted in FIG. 5. The fourth panel section **48** and the fifth panel section **50** together define a mounting structure that can interact with one of the cover **12** or the case **110** to form a rear support for the mobile device **10**. In some examples, the fourth panel section **48** is configured to remain approximately coplanar with the first panel section **32**, while the fifth panel section **50** is folded along the fourth fold line **54**. The fifth panel section **50** extends upwardly and inwardly away from the fourth panel section **48** and/or the first panel section **32** to form an acute angle between the fifth panel section **50** and a rear surface of the cover **12** or case **110** of the mobile device **10**. In some examples, a notch or groove is formed within the rear surface of the cover **12** or case **110** to catch and secure the fifth panel section **50**, which can then create a readily adjustable releasable coupling between the input board **30** and the mobile device **10**. In some examples, a series of notches are formed within the rear surface of the cover **12** or case **110** so as to allow the

fifth panel section **50** to be repositioned relative to the mobile device **10**, which in turn adjusts the angle of the mobile device **10** (and display **14**) relative to the first panel section **32** and other inputs on the input board **30**.

(29) Alternatively, one or more stands **62** coupled to the cover **12** or the case **110** can be used to support the mobile device **10** for use with the input board **30**. As depicted in FIG. 7, a stand **62** extends downwardly and outwardly away from a rear surface of the case **110**. In some examples, the stands **62** are hingedly coupled to the mobile device **10** or case **110**. The stands **62** can be manually rotated outward, away from the mobile device **10** to a variety of different positions. In some examples, a hinge (not shown) supporting each stand **62** is configured to adjust an angle of the mobile device **10** relative to a surface below. For example, the mobile device **10** can be rotated between an angle of between about 95 degrees and about 175 degrees relative to the ground surface below. In some examples, the stand **62** is formed as a part of a push release mechanism, such that pushing downward (e.g., inward) on one side of the stand **62** causes the stand **62** to release and extend outward from the mobile device **10** or case **110**, where it can then be readily manipulated and rotated by a user to adjust the mobile device **10** to a desired orientation. The stand **62** can operate independently of the input board **30**, such that the stand **62** can be deployed regardless of the location of the input board **30**. In other examples, the stand **62** is configured to automatically deploy upon removal or deployment of the input board **30**. In still other examples, the one or more stands **62** are positioned within the base wall **114** and can be accessed and deployed once the input board **30** has been rotated out of the storage position.

(30) In still other examples, the coupling between the input board **30** and the mobile device **10** can be used to prop up or support the mobile device **10** at an angle relative to the surface below and at a preferred or desired angle relative to the input board **30**. The input board **30** can be coupled to the mobile device **10** and/or the case **110** using a hinge **64**. The hinge **64** can be formed within a pocket or recess formed within a rear wall of the cover **12** or case **110**. In some examples, the hinge extends along a full width of the first panel section **32**. The hinge **64** can be configured to resist but not prohibit relative motion between the input board **30** and the mobile device **10**. In some examples, the resistance within the hinge **64** can be used to position the mobile device **10** relative to the input board **30**. Accordingly, a user can adjust the angle between the input board **30** and the mobile device **10** by moving one or both of the input board **30** and mobile device **10** about an axis defined by the hinge **64**. The angle between the input board **30** and the mobile device **10** can be adjusted over a range of about 270 degrees, between a position where the input board **30** is stowed alongside the cover **12** or case **110** and a position where the input board **30** is deployed and the angle between the input board **30** and the mobile device **10** is about 90 degrees. The range of motion can be adjusted depending upon the storage position of the input board **30**. In some examples, the hinge **64** includes a lock that is configured to secure a relative position of the input board **30** and mobile device **10**. The lock on the hinge **64** can secure the input board **30** and mobile device **10** in several discrete and predetermined orientations. In some examples, the input board **30** includes one or more weights positioned at distal ends of the panel sections **32**, **34**, **36** to operate as counterweights that oppose tipping of the mobile device **10**.

(31) As discussed above, the input board **30** can be stored relative to the mobile device **10** in a variety of different locations. For example, the input board **30** can be stored within the cover **12** of the mobile device **10**. To promote easy storage and access, a recess **70** is formed within the back side of the rear wall **16**. As depicted in FIG. 5, the recess **70** is formed from a series of walls **72** extending inwardly from the rear wall **16**, toward the display **14** and front of the mobile device **10**. The recess **70** can be sized to form an interference fit with the folded input board **30**, for example, so that friction will retain the input board **30** within the recess **70** in the stored position. The recess **70** can be sized so that a depth of the recess is greater than or equal to a total thickness of the input board **30** in the folded position. Accordingly, when the input board **30** is in the folded and stowed position, the input board **30** is entirely received within the recess so that the entire input board is

positioned inward from an outermost (e.g., the rear) surface of the rear wall **16**. In some examples, a portion of the input board **30** extends approximately flush (e.g., ± 0.25 cm) with the rear wall **16** of the cover so as to provide a smooth, continuous rear surface to the mobile device **10**. In some examples, a lip can extend around a portion of a perimeter of the recess **70** to further secure the input board **30** within the recess **70**. When the input board **30** is formed of a pliable material, the input board **30** can flex as it contacts the lip. Once beyond the lip, the resilient material of the input board **30** will allow the input board **30** to return to its normal shape, where it can then be stored indefinitely. In still other examples, the input board **30** is removably received within the recess **70** using a magnetic coupling.

(32) In some examples, the recess **70** can house or otherwise include the hinge **64**. The hinge **64** can be secured to one of the walls **72** or to the floor **74** of the recess **70**. The hinge **64** can extend a portion of the length (e.g., about 70%-90%) of the mobile device **10**, which corresponds to the length of the recess **70**. In some examples, the hinge **64** is defined by a thickness that corresponds to approximately the thickness of the input board **30** in the folded position. As discussed above, the hinge **64** defines an axis of rotation for the input board **30** to rotate, relative to the mobile device **10**.

(33) The hinge **64** can be arranged to restrict the amount and/or rate of relative motion between the input board **30** and the mobile device **10**. In some examples, the hinge **64** includes a pin that forms an interference fit between the input board **30** and the cover **12**. The interference fit resists but does not prevent relative motion between the input board **30** and the cover **12**, such that without additional actuation (e.g., rotational force), the relative angle between the input board **30** and display **14** of the mobile device **10** will be maintained. In some examples, the hinge **64** is a compound joint, including at least two pins that allow the hinge **64** to rotate about multiple parallel axes. In such examples, a first hinge joint within the hinge **64** can be used to rotate the input board **30** into and out of the recess **70**, while the second hinge joint can be used to rotate the first hinge joint and the input board **30** about a second axis that allows the input board **30** to rotate around the mobile device **10**. In still other examples, the input board **30** is both axially and rotatably movable using the hinge **64**. For example, the input board **30** can translate axially away from the pin of the hinge **64** before rotating relative to the mobile device **10** to a deployed position. Similarly, the input board **30** can be stowed by rotating the input board **30** toward the cover **12**, sliding the input board inward, toward the pin of the hinge **70**, and then finishing the rotation of the input board **30** about the hinge **64** until the input board is positioned entirely within the recess **70**. In some examples, the hinge **64** can include a lock that can secure the hinge **64** in different positions. For example, the lock can be configured to secure the input board within the recess **70** in the stowed position, until an actuator (e.g., a mechanical or electrical actuator) is depressed or otherwise manipulated. Although described with respect to the cover **12**, the hinge **64** can also be included within the case **110**.

(34) In other embodiments, and as depicted in FIG. 7, the mobile device **10** is received within an accessory case, shown as the case **110**. The case **110** generally includes a housing **112** that has a base wall **114** and a perimeter wall **116**. The perimeter wall **116** extends away from the base wall. The perimeter wall **116** and the base wall **114** together define a cavity **118** that is sized to receive a mobile device, such as the mobile device **10** or a tablet, for example. In some examples, the base wall **114** and/or the perimeter wall includes a recess that is configured to store a stylus.

(35) The housing **112** is designed to receive and store an input board, like the input board **30**. In some examples, a channel **120** is formed within the base wall **114**. The channel **120** is defined by two or more walls **122** that extend inward from a first perimeter side of the base **114**. For example, and as shown in FIG. 7, the channel **120** is defined by four rectangular outer walls **122** and channel floor **124**. The rectangular walls **122** extend inward, across the base wall **114** but not entirely through the base wall **114**. The rectangular walls **122** and channel floor **124** together define a pocket **126** within the base wall **114** that is separated from the cavity **118**. By extending inwardly

from one side, but not through the entire base wall **114**, the pocket **126** can receive and secure the input board **30** in a way that restricts contaminants from contacting or otherwise damaging the input board **30**. Similarly, the input board **30** can be substantially hidden when not in use. In some examples, a depth of the pocket **126** is chosen to be less than a length of the input board **30** in the folded configuration so that a portion of the input board **30** extends outwardly from the case **110** for easy removal. In other examples, the depth of the pocket **126** is chosen to be greater than the length of the input board **30** in the folded configuration so that the entire input board **30** is received within the case **110** when the input board **30** is not in use. The width of the channel **120** is greater than a thickness of the input board **30** so that the input board **30** can be pushed into the pocket **126**. The depth of the pocket **126** is greater than a depth of the cavity **118**.

(36) The input board **30** is removably received within the pocket **126**, and can be selectively electrically coupled to the mobile device **10** and/or a battery onboard the case **110**. In some examples, the input board **30** includes an onboard and rechargeable battery that receives electrical power from the battery **22** of the mobile device **10**. To transmit electrical power between the battery **22** of the mobile device **10** and the battery of the input board **30**, the input board **30** is selectively coupled to the rechargeable battery **22** through terminals **128** that are coupled with the rechargeable battery **22**. The terminals **128** can be formed within the pocket **126** (e.g., within the channel floor **124**), for example, so that an electrical connection is formed between the input board **30** and the terminals **128** whenever the input board **30** is received within the pocket **126**. Accordingly, the input board **30** can be recharged whenever the input board **30** is stored and not in use. In such examples, the input board **30** can be configured to communicate with the mobile device **10** over a wireless frequency (e.g., Bluetooth, etc.) to transmit input-based commands. In still other examples, the pocket **126** is configured to operate as a wireless charger for the input board **30**, so that energy is transmitted from the rechargeable battery **22** to the input board **30** for later use. Alternatively, the pocket **126** can be arranged so that whenever the input board **30** is received within the pocket **126**, the input board **30** is disconnected from the battery **22**.

(37) The battery **22** can be placed in communication with the input board **30** in both the storage position and the deployed position. For example, the case **110** can support a series of externally accessible terminals **130** that extend through one or more of the perimeter wall **116** and/or the base wall **114**. The terminals **130** can be in direct electrical communication with the battery **22** through a coupling with the externally accessible terminals **56**, for example. Like the externally accessible terminals **56**, the terminals **130** can include both power terminals and data terminals that are configured to selectively engage with the data terminals **58** and power terminals **60** of the input board **30**. When the terminals **130** are placed into communication with the data terminals **58** and power terminals **60** of the input board **30**, the input board **30** is placed in electrical communication with the battery **22** of the mobile device **10**, and is configured to transmit signals associated with selected inputs between the mobile device **10** and the input board **30** and electrical power between the battery **22** and the input board **30**.

(38) The input board **30** can be deployed from the case **110** in several of the same ways that the input board **30** can be deployed from the cover **12** of the mobile device **10**. For example, the input board **30** can be removed from the pocket **126**, placed into electrical communication with the terminals **130**, and then folded into the deployed position by rotating the panel sections **34**, **36**, **48**, **50** about the different fold lines **42**, **40**, **52**, **54** formed on the input board **30**. In the fully deployed position and configuration shown in FIG. **4**, the width (e.g., lateral distance) of the input board **30** is at least 1.5 times larger than a length (e.g., the lateral distance when rotated 90 degrees) of the mobile device **10**. In some examples, the input board **30** can be pulled out from the pocket **126** by hand. In other examples, a mechanical or electrical actuator can be formed within the case **110** or cover **12** to eject the input board **30**. In still other examples, the pocket **126** can be formed within the base wall **114** only, and the input board **30** is hingedly coupled to the case **110** and rotatable between the stowed and deployed positions. Once the desired angle relative to the mobile device **10**

is reached, the panel sections **32, 34, 36** of the input board **30** can be unfolded so that the input board **30** is ready for use.

(39) In some examples, the input board **30** can form the rear of the case **110**. In such embodiments, the input board **30** can be formed of a rigid, semi-rigid, or flexible and resilient material that can be used to protect the mobile device **10** from damage in the event of a drop or other impact. The input board **30** can be locked into the folded and stowed position using a magnetic or otherwise mechanical lock. The lock can be electronically unlocked by the mobile device **10** or through a release button or snap locking mechanism. When the input board **30** forms the rear of the case **110**, the input board **30** can extend along the base wall **114** of the case **110** or can be embedded within a portion of the base wall **114**, and the input board **30** extends flush with the case **110** when stowed, and the base wall **114** completely surrounds the input board **30**. The input board **30** can be locked to the base wall **114**, either within a center of the base wall **114** or to a lip formed within the base wall **114** that surrounds the input board **30**. In examples where the input board **30** forms the rear of the case **110**, the hinge (e.g., the hinge **64**) is secured to the case **110**. The input board **30** is deployed by rotating the input board **30** around the stationary hinge and to the front of the mobile device **10**, then unfolding the panel sections **32, 34, 36**. In other examples, the hinge **64** is coupled directly to the input board **30**, and can be locked into place relative to the case **110** once the input board **30** is removed from the case **110** for use. Accordingly, the hinge **64** moves with the input board **30**. The case **110** can include a track or channel that receives and constrains the relative motion of the hinge **64** and input board **30** relative to the case **110**.

(40) Referring now to FIG. **8**, another exemplary embodiment of an input board **140** is provided. Like the input board **30**, the input board **140** is formed of a series of foldable panel sections **142, 144, 146, 152**. Like the input board **30**, the input board **140** generally includes three panel sections **142, 144, 146** that support a series of alphanumeric keys **40**. In some examples, the alphanumeric keys **40** include both a traditional keyboard and a keypad. The alphanumeric keys **40** can extend across one or more of the panel sections **142, 144, 146**. The panel sections **144, 146, 152** are hingedly coupled to the first panel section **142** along a series of fold lines **148, 150, 154**. As depicted, the fold line **154** extends perpendicular to the fold lines **148, 150**.

(41) The input board **140** includes a track pad **42** that extends along the fourth panel section **152**. Like the input board **30**, the input board **140** can be arranged to fold between a fully deployed position (shown in FIG. **8**) and a stowed position where the panel sections **142, 144, 146, 152** are stacked in a folded or hinged assembly. In some examples, there is a dedicated or designed folding order to return the panel sections **142, 144, 146, 152** to the folded and stowed position. In the embodiment depicted in FIG. **8**, the panel section **152** is arranged to be folded along a top of the panel sections **144, 146**, which in turn are folded onto and extend across the first panel section **142**. In some examples, the panel section **152** is narrower than the panel section **142**, and the panel section **152** is centered relative to the panel section **142**. Alternatively, the panel section **152** can extend along the entire width of the first panel section **142**. In some examples, the input board **140** includes magnetic or otherwise mechanical locks that can be used to selectively secure the input board **140** in the folded or stowed position. The panel section **152** can include a variety of other inputs as well, including volume control, brightness control, music controls, and/or numbers, for example.

(42) Using the above described mobile devices **10**, cases **110**, and input boards **30, 140** significantly more efficient data entry processes can be accomplished. Word processing software and other data entry programs typically reserved to traditional computers and laptops can be included on cell phones, tablets, and other mobile devices, as these software programs will be useful with devices that previously were difficult to quickly or accurately enter significant amounts of data. Additionally, the various input boards **30** described herein are highly portable and easily concealed to protect and transport the input boards **30** for rapid use.

(43) While not specifically shown in the FIGURES, various other modifications to the input board

30 and its deployment mechanisms to promote fast and efficient use. For example, in the storage position or configuration, the input board **30** can face inward (e.g., toward the display **14**) or outward (toward the rear of the mobile device **10**). The input board **30** can include a swivel joint that allows the input board **30** to first rotate 180 degrees to the opposite orientation before rotating the input board to the deployed position. The hinge **64** can lock or otherwise prohibit rotation of the input board **30** until the orientation of the input board **30** reaches the appropriate direction. In some examples, the input board **30** can be stored in either orientation. As discussed above, the input board **30** can operate using a wired or wireless connection to the mobile device **10**. In instances where the input board **30** is hingedly coupled to the mobile device **10** or case **110**, a wired coupling can be maintained at all times, regardless of the orientation of the input board **30**. Additionally, the deployment of the input board **30** can include rotation and/or translation about multiple axes. For example, the input board **30** may first be urged rearward, out of the recess **70**, before it can be rotated about the hinge **64**.

(44) Various configurations of the input board **30** are possible as well. Although described as foldable panel segments, the input board **30** can be formed from a series of selectively coupled panel segments. The coupling can be magnetic, for example, which allows for a fast, easy, and consistent assembly process. In some examples, inputs extend across more than one panel section. For example, in some embodiments, the track pad **42** extends on each of the first panel section **32** and the second panel section **34**. The different panel sections can be formed of a resilient material that is configured to flex under stress.

(45) In some examples, the mobile device is configured with two software operating interfaces or systems that may adjust the functionality and program availability depending on whether the input board **30** is deployed or stored. When the input board **30** is stored, the mobile device **10** can operate in its limited and normal operating capacity. Once the input board **30** transitions to the deployed position, however, a processor within the mobile device **10** can transition the mobile device to the more robust software operating interface, and additional programs (e.g., word processing programs, spreadsheet programs, etc.) can be unlocked. The two software operating interfaces are configured to communicate with one another through a common master operating system (e.g., Apple iOS, Microsoft, Android, etc.)

(46) Although this description may discuss a specific order of method steps, the order of the steps may differ from what is outlined. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(47) As utilized herein, the terms “approximately”, “about”, “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

(48) It should be noted that the term “exemplary” as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(49) The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two

members directly or indirectly to one another. Such joining may be stationary (e.g., permanent, etc.) or moveable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

(50) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” “between,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(51) It is important to note that the construction and arrangement of the mobile device accessory as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. It should be noted that the elements and/or assemblies of the components described herein may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and other exemplary embodiments without departing from scope of the present disclosure or from the spirit of the appended claims.

Claims

1. An accessory case for a mobile device, comprising: a housing having a base wall and a perimeter wall extending away from the base wall, the perimeter wall and base wall together defining a cavity to receive a mobile device therein; a channel formed within the base wall, the channel being defined by a plurality of walls extending inward from a first perimeter side of the base wall or, toward but not through a second perimeter side of the base wall, wherein the channel defines a pocket within the base wall separated from the cavity; and an input board received within the pocket, the input board being formed by: a first panel section, a second panel section connected to the first panel section along a first fold line, a third panel section connected to the first panel section along a second fold line extending approximately parallel to the first fold line, and a fourth panel section connected to the first panel section along a third fold line extending approximately perpendicular to the first fold line, wherein each of the panel sections includes at least one input, wherein the input board is configured to transition between a folded position and an unfolded position, wherein in the folded position, a width of the input board is less than a width of the channel, wherein in the unfolded position, the width of the input board is greater than the width of the channel, and wherein in the folded position, the second panel section and the third panel section extend along a front side of the first panel section and the fourth panel section extends along a back side of the first panel section.
2. The accessory case of claim 1, wherein a depth of the cavity is greater than a width of the pocket, and wherein the input board is defined by a thickness less than the width of the pocket in both the folded position and the unfolded position.
3. The accessory case of claim 1, wherein the input board is selectively electrically coupled to and powered by a battery of the mobile device, wherein when the input board is received within the pocket, the input board is decoupled from the battery of the mobile device.

4. The accessory case of claim 3, wherein an electrical coupling extends from the battery of the mobile device and through at least one of the perimeter wall or the base wall to form an externally accessible coupling that can be selectively contacted by the input board to create an electrical connection between the battery and the input board.
 5. The accessory case of claim 1, wherein in the folded position, the second panel section and the third panel section extend along the first panel section.
 6. The accessory case of claim 1, wherein a perimeter of the first panel section is approximately twice a perimeter of the second panel section, and wherein in the folded position, the second panel section extends in a plane offset from the first panel section, and wherein in the unfolded position, the second panel section extends coplanar with the first panel section.
 7. The accessory case of claim 1, wherein the input board further comprises a fifth panel section, the fifth panel section extending away from the fourth panel section and connected to the fourth panel section along a fourth fold line, the fourth fold line extending parallel to the third fold line, wherein the fourth and fifth panels do not have inputs received thereon.
 8. The accessory case of claim 1, wherein the first perimeter side of the base wall is a rear side of the base wall and the second perimeter side of the base wall is a front side of the base wall, and wherein in the folded position, the input board defines at least a portion of an outermost surface of the accessory case, and wherein the input board is hingedly coupled to the base wall.
 9. The accessory case of claim 1, wherein the input board includes at least one track pad.
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